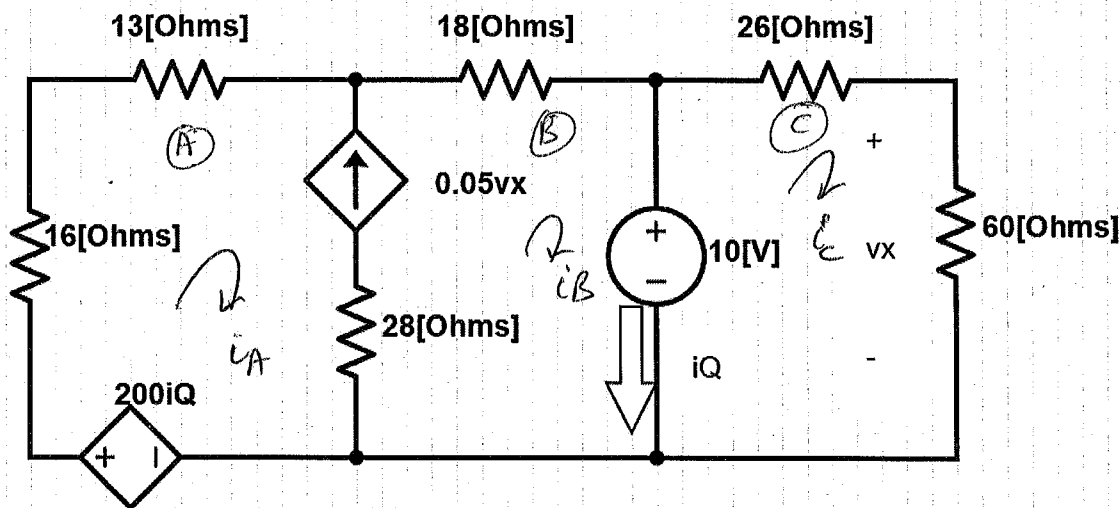


Name: Solutions (please print)

ECE 2202 - Quiz 1
February 9, 2026

Problem 1 of 2:

Solve the following circuit using **Mesh Current Method**. What is i_Q and v_x ?



Unknowns:
 $i_A, i_B, i_C,$
 i_Q, v_x

(Mesh A) + (Mesh B) supermesh
 $-200i_Q + i_A 16[\Omega] + i_A 13[\Omega] + i_B 18[\Omega] + 10[V] = 0$

(Constraint)
 $i_B - i_A = 0.05v_x$

(Mesh C)
 $i_C 26[\Omega] + i_C 60[\Omega] - 10[V] = 0$

(i_Q) $i_Q = i_B - i_C$

(v_x) $v_x = 60[\Omega] i_C$

Solve: $i_C = \frac{10}{26+60} = 0.1163[A]$

$v_x = 60(0.1163) = 6.9767[V]$

$$\begin{bmatrix} 16+13 & 18 & -200 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} i_A \\ i_B \\ i_Q \end{bmatrix} = \begin{bmatrix} -10 \\ 0.05(6.9767) \\ -0.1163 \end{bmatrix}$$

$i_A = -0.19757[A]$

$i_B = 0.15127[A]$

$i_Q = 0.03477[A]$

Problem 2 of 2:

Using **Mesh Current Method** only on the circuit below:

- a) How many equations are needed to completely solve the circuit?
- b) Write the complete system of independent equations that could be used to solve the circuit. You do not need to solve the system of equations. Be sure to include units on all terms in the equations. Be sure to label the circuit with all variables that appear in the equations.

$$\textcircled{N_4} \quad -N_4 + (E - iR) 39(\Omega) + 5(V) + 6E 22(\Omega) = 0$$

